

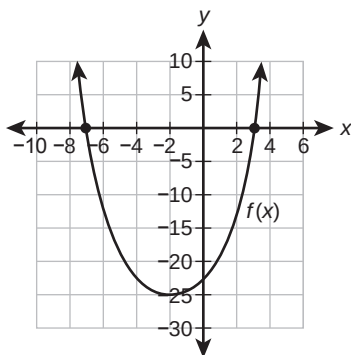
Graphing Polynomials

Because any root of $f(x)$ is also an x -intercept of the graph of $f(x)$, you can determine the roots of a polynomial function from its graph.

Using a Graph to Determine the Roots of a Polynomial

EXAMPLE 1

A Find the roots of $f(x) = x^2 + 4x - 21$.



SOLUTION

The graph of $f(x) = x^2 + 4x - 21$ appears to intersect the x -axis at $(3, 0)$ and $(-7, 0)$, which means that 3 and -7 seem to be roots, both of which are consistent with the possible rational roots according to the rational root theorem.

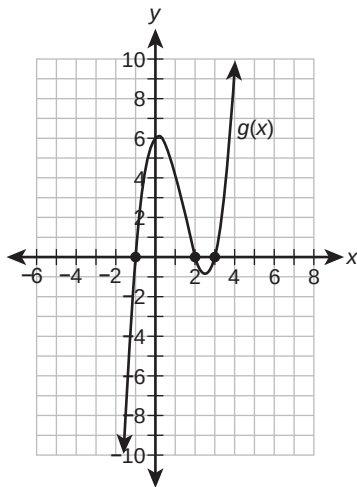
CHECK

Find $f(3)$ and $f(-7)$ to check your answer.

$$\begin{aligned} f(3) &= 3^2 + 4 \cdot 3 - 21 & f(-7) &= (-7)^2 + 4 \cdot (-7) - 21 \\ &= 9 + 12 - 21 & &= 49 - 28 - 21 \\ &= 0 \checkmark & &= 0 \checkmark \end{aligned}$$

Since $f(3) = 0$ and $f(-7) = 0$, 3 and -7 are the roots of $f(x)$.

B Find the roots of $g(x) = x^3 - 4x^2 + x + 6 = 0$.

**SOLUTION**

The graph of $g(x) = x^3 - 4x^2 + x + 6 = 0$ appears to intersect the x -axis at $(-1, 0)$, $(2, 0)$, and $(3, 0)$, which means that -1 , 2 , and 3 seem to be roots, all of which are consistent with the possible rational roots according to the rational root theorem.

CHECK

You can check your answer by factoring. Use any one of the roots obtained from the graph and turn it into a factor: $x - (-1) = x + 1$. Use synthetic division to divide.

$$\begin{array}{r|rrrr} -1 & 1 & -4 & 1 & 6 \\ & & -1 & 5 & -6 \\ \hline & 1 & -5 & 6 & 0 \end{array}$$

Use the quotient to write the polynomial as a product of factors.

$$\begin{aligned}x^3 - 4x^2 + x + 6 &= (x + 1)(x^2 - 5x + 6) \\ &= (x + 1)(x - 2)(x - 3) \checkmark\end{aligned}$$

The factors show that the roots are -1 , 2 , and 3 . ■

Graphing a Polynomial Function by Using Intercepts

You can draw the graph of a function by finding its x -intercepts and incorporating what you know about the end behavior of power functions.

EXAMPLE 2

Draw the graph of $f(x) = x^4 - 3x^3 - 11x^2 + 3x + 10$.

SOLUTION

Step 1 Use the rational root theorem to find the possible rational roots of $f(x)$. The leading coefficient is 1 and the constant term is 10, so the possible rational roots are ± 1 , ± 2 , ± 5 , and ± 10 .

Step 2 Use synthetic division to test the possible roots.

$$\begin{array}{r|rrrrrr} 1 & 1 & -3 & -11 & 3 & 10 \\ & & & 1 & -2 & -13 & -10 \\ \hline & 1 & -2 & -13 & -10 & 0 \end{array}$$

Since 1 is a root, $x - 1$ is a factor. You can write the polynomial as $(x - 1)(x^3 - 2x^2 - 13x - 10)$.

Use synthetic division to find a factor of the cubic polynomial $x^3 - 2x^2 - 13x - 10$. The leading coefficient is 1 and the constant term is -10 , so the possible rational roots are ± 1 , ± 2 , ± 5 , and ± 10 .

$$\begin{array}{r|rrrr} -1 & 1 & -2 & -13 & -10 \\ & & -1 & 3 & 10 \\ \hline & 1 & -3 & -10 & 0 \end{array}$$

Since -1 is a root, $x + 1$ is a factor. You can write the polynomial as $(x - 1)(x + 1)(x^2 - 3x - 10)$.

Step 3 You could continue to use synthetic division to find the factors of the quadratic polynomial, but factoring it into two binomials is quicker. The function in factored form is $f(x) = (x - 1)(x + 1)(x - 5)(x + 2)$. The x -intercepts, then, are 1, -1 , 5, and -2 . Plot the x -intercepts.

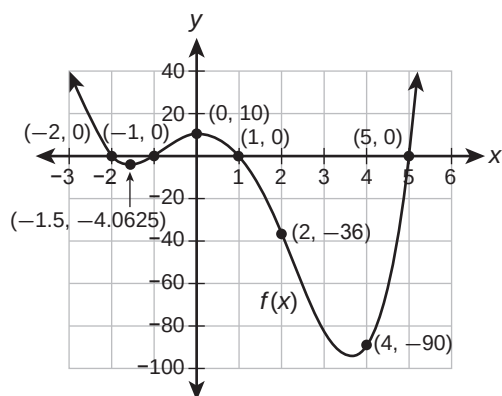
► **Think About It** Note that there are an infinite number of polynomials with these four roots. These polynomials all have the form

$$f(x) = a(x - 1)(x + 1)(x - 5)(x + 2).$$

When you find another point, you can determine a by substituting for x and $f(x)$ in the equation.

Step 4 Graph the polynomial. Because the function is a polynomial with even degree and a positive leading coefficient, the graph approaches positive infinity as x approaches both negative and positive infinity. Use synthetic division to find additional points.

	1	-3	-11	3	10
-1.5	1	-4.5	-4.25	9.375	-4.0625
0	1	-3	-11	3	10
2	1	-1	-3	-23	-36
4	1	1	-7	-25	-90



Graphing a Polynomial Function by Using Roots

EXAMPLE 3

Draw the graph of $p(x) = x^3 - x^2 - 5x + 2$.

SOLUTION

Step 1 Use the rational root theorem to find the possible rational roots of $p(x)$. The leading coefficient is 1 and the constant term is 2, so the possible rational roots are ± 1 and ± 2 .

Step 2 Test the possible rational roots until you find a root of the polynomial.

	1	-1	-5	2
-1	1	-2	-3	5
1	1	0	-5	-3
-2	1	-3	1	0

Since -2 is a root, $x + 2$ is a factor. You can write the polynomial as $(x + 2)(x^2 - 3x + 1)$.

Step 3 Use the quadratic formula to solve $x^2 - 3x + 1 = 0$ and find the remaining two irrational roots.

$$\begin{aligned}x &= \frac{3 \pm \sqrt{(-3)^2 - 4 \cdot 1 \cdot 1}}{2 \cdot 1} \\&= \frac{3 \pm \sqrt{5}}{2}\end{aligned}$$

The roots of the polynomial are -2 , $\frac{3 + \sqrt{5}}{2}$, and $\frac{3 - \sqrt{5}}{2}$.

Step 4 Graph the function. Plot the x -intercepts, which are -2 ,

$$\frac{3 + \sqrt{5}}{2} \approx 2.62, \text{ and } \frac{3 - \sqrt{5}}{2} \approx 0.38.$$

The function is cubic, so it is S-shaped. Because the function is a polynomial with odd degree and has a positive leading coefficient, the graph approaches negative infinity as x approaches negative infinity, and the graph approaches positive infinity as x approaches positive infinity.

Find and plot a few additional points.

x	-3	-1	0	1
$p(x)$	-19	5	2	-3

